GEOMETRY · TRIANGLE RELATIONSHIPS

# Triangle Explorer

Explore & Reflect

Name _____ Date _____ Period _____

Open **Triangle Explorer** on Mr. K's Classroom: Geometry → Triangle Relationships → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

## 1 Predict before you touch the tool

1. Do you think the angle sum changes when a triangle becomes obtuse?

_____

_____

## 2 Explore use the tool and record what you see

### Angles (A(–5, –3), B(5, –3), and C as given)

Drag C to each point. Turn on the exterior angle at C.

| C | ∠A | ∠B | ∠C | Sum | Type | Exterior angle at C |
|---|---|---|---|---|---|---|
| (1, 5) | | | | | | |
| (5, 5) | | | | | | |
| (–8, –1) | | | | | | |

### Triangle centers (use the obtuse triangle)

| Center | Where these lines meet | Inside or outside? |
|---|---|---|
| Centroid | | |
| Incenter | | |
| Circumcenter | | |
| Orthocenter | | |

### Can it be a triangle?

Use the second tab.

| Side lengths | Two shortest add to | Triangle? |
|---|---|---|
| 3, 4, 6 | | |
| 3, 4, 8 | | |
| 5, 5, 10 | | |
| 4, 6, 9 | | |

## 3 Write the rule fill in each blank from what you observed

1. The angles of every triangle add up to _____.

2. An exterior angle equals the sum of the two _____ interior angles.

3. The longest side is across from the _____ angle.

4. The sum of any two sides must be _____ the third side.

5. The circumcenter is the same distance from all three _____. The incenter is the same distance from all three _____.

## 4 Check no tool: show your work

1. Two angles are 47° and 68°. Find the third.

2. The exterior angle at C is 115° and ∠A = 40°. Find ∠B.

3. Two sides are 5 and 9. What lengths can the third side have?

4. ∠A = 50°, ∠B = 70°, ∠C = 60°. List the sides from shortest to longest.

## 5 Reflect

Why can't 3, 4 and 8 make a triangle?